

Functional sequences, series and spaces – 5

Hilbert spaces and orthogonal systems

Homework

13 сентября 2026 года

Theoretical minimum. In a Hilbert space the best approximation by a closed subspace is the orthogonal projection, and orthogonality replaces every computation by a Pythagorean identity. Bessel's inequality holds for any orthonormal system, Parseval's identity exactly for complete ones, and Riesz–Fischer says that any ℓ^2 sequence of coefficients is realised. Gram–Schmidt with a weight produces orthogonal polynomials, which satisfy a three-term recurrence and have simple interlacing roots; the classical weights give Legendre, Chebyshev and Hermite. An orthonormal sequence tends weakly to zero but keeps norm one, so the unit ball is not compact and weak convergence plus convergence of norms is what upgrades to convergence in norm. Completeness of a system is usually proved by showing that a vector orthogonal to all of it must vanish — which is a moment problem in disguise.

1. (Капачёв 1.89) Let (f_n) be a uniformly bounded sequence of functions on $[0, 1]$ which is orthonormal: $\int_0^1 f_n f_m = \delta_{nm}$. Can the sequence $f_n(x)$ have a pointwise limit at every point?
2. (Капачёв 1.92) Is it true that the sum $V + W$ of two closed linear subspaces of a Hilbert space is closed? Prove it or construct a counterexample.
3. (S.T. Yau CSMC 2018) Let (f_n) in $L^2(\mathbb{R}^d)$ satisfy $\|f_n\|_{L^2} = 1$.
 - a) Show that some subsequence converges weakly to a function f , that is $\langle f_{n_j}, g \rangle \rightarrow \langle f, g \rangle$ for every $g \in L^2(\mathbb{R}^d)$.
 - b) Show that if $f_n \rightarrow f$ weakly and $\|f_n\| \rightarrow \|f\|$, then $\|f_n - f\| \rightarrow 0$.

4. Let (e_n) be an orthonormal system in a Hilbert space and let x be a vector. Prove that $\sum_n |\langle x, e_n \rangle|$ may diverge while $\sum_n |\langle x, e_n \rangle|^2$ always converges, and characterise the sequences (c_n) for which $\sum c_n \langle x, e_n \rangle$ converges for every x . (The values of $\zeta(2)$ and $\zeta(4)$ from Parseval are computed in Trig-6.)
5. Prove that vectors $x_1, \dots, x_n$ are linearly independent if and only if their Gram determinant is non-zero, and that the squared distance from x to their span equals the ratio $\frac{G(x_1, \dots, x_n, x)}{G(x_1, \dots, x_n)}$ (compare with Linear algebra – 5).
6. Prove that the Chebyshev polynomials are orthogonal on $[-1, 1]$ with the weight $(1-x^2)^{-1/2}$, and explain the difference between their extremal property in the uniform norm (Continuous functions – 9) and the extremal property of the Legendre polynomials in L^2 .
7. Prove that ℓ^2 is complete and separable, that its closed unit ball is not compact, and that every orthonormal sequence converges weakly to zero.
8. Prove the Riesz–Fischer theorem, and prove that Bessel’s inequality becomes an equality exactly when the vector lies in the closed span of the system.
9. Prove that the Hermite polynomials, obtained from $1, x, x^2, \dots$ by Gram–Schmidt with the weight e^{-x^2} on $\mathbb{R}$, satisfy $H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}$ up to normalisation, and compute their norms in terms of the gamma function.
10. Prove that a Hilbert space with a countable dense subset possesses a countable orthonormal basis, and deduce that all infinite-dimensional separable Hilbert spaces are isometrically isomorphic to ℓ^2 .